Laurent Series

We know that if a function is analytic everywhere on an open disk that we have a Taylor series on that disk. It is a shame that even just one point of non-analyicity prevent us from having series expansion everywhere else. It turns out that there is a type of series which allows us to still expand our function in powers of $(z - z_0)$ even if we have point where the function is not analytic.

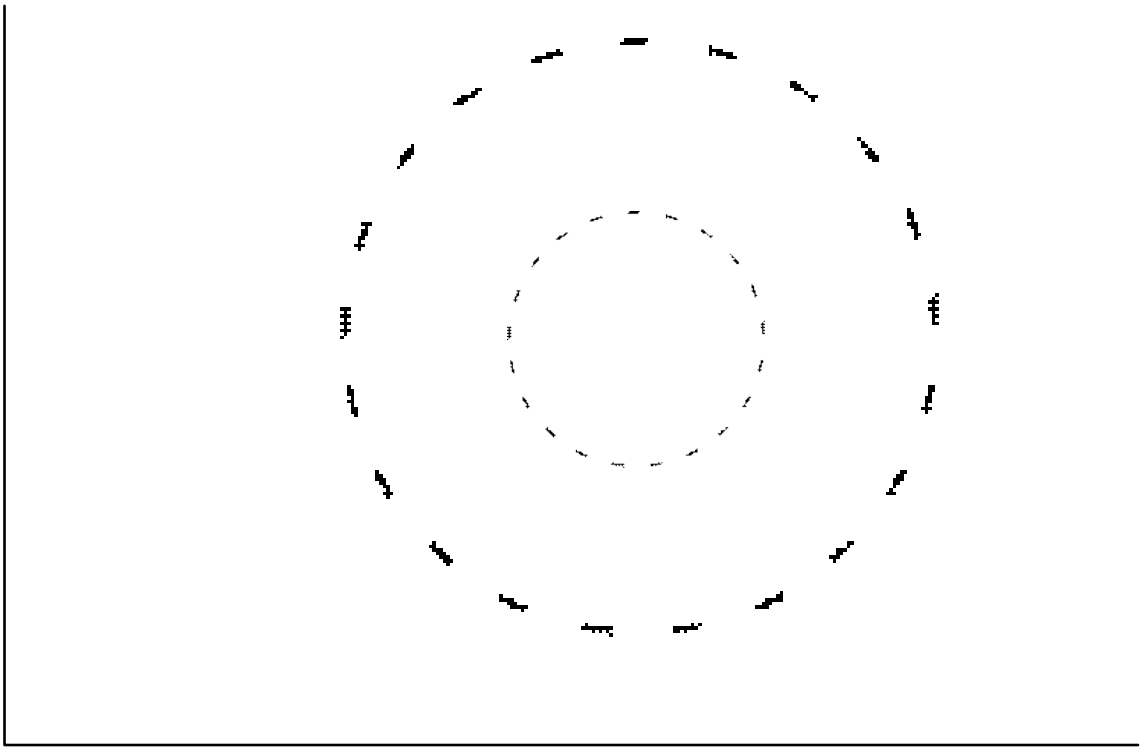
Laurent's Theorem: Suppose that a function f is analytic throughout an annular domain $R_1 < |z - z_0| < R_2$, centered at a point z_0, and let C be any positively oriented simple closed contour around z_0 lying in the annular region. Then at each point in the annular domain $f(z)$ has the series representation.

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0) + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \qquad R_1 < |z - z_0| < R_2$$

Where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad n = 0,1,2, \ldots$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad n = 1,2, \ldots.$$



Laurent Series

Another way Luarent series are represented is to allow negative values for n in which case our theorem becomes:

Laurent's Theorem: Suppose that a function f is analytic throughout an annular domain $R_1 < |z - z_0| < R_2$, centered at a point z_0, and let C be any positively oriented simple closed contour around z_0 lying in the annular region. Then at each point in the annular domain $f(z)$ has the series representation.

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \qquad R_1 < |z - z_0| < R_2$$

$$c_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad n = 0, \pm 1, \pm 2, \pm 3$$

We can show this by interchanging $-n$ with n in our expression for b_n $\quad b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad n = 1,2, \ldots.$

Then $b_{-n} =$

And we can then write:

$$\sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} = \sum_{n=-\infty}^{-1} b_{-n} (z - z_0)^n$$

Then

$$f(z) = \sum_{n=-\infty}^{-1} b_{-n} (z - z_0)^n + \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Or just that

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \qquad R_1 < |z - z_0| < R_2$$

$$c_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad n = 0, \pm 1, \pm 2, \pm 3$$

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Proof!

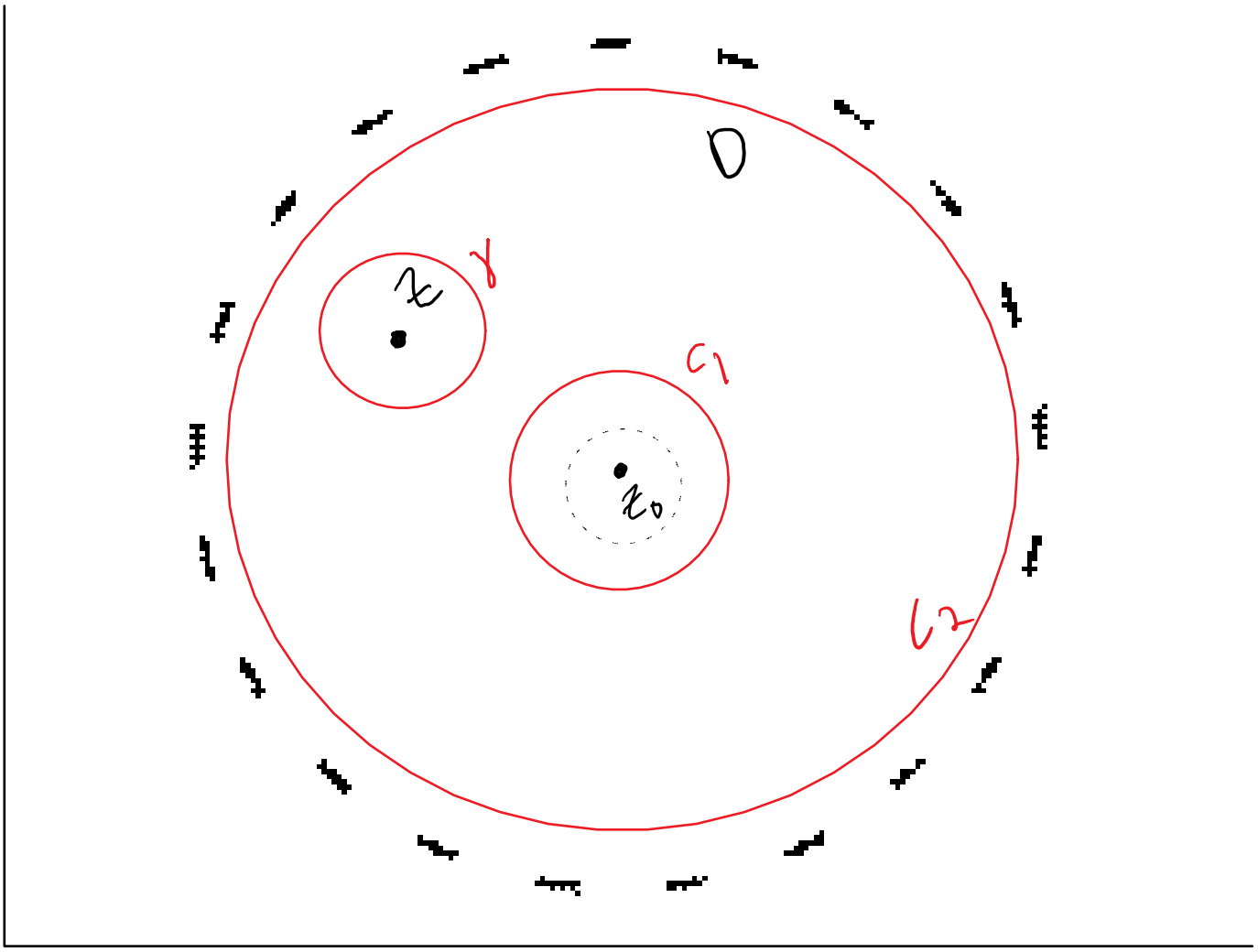
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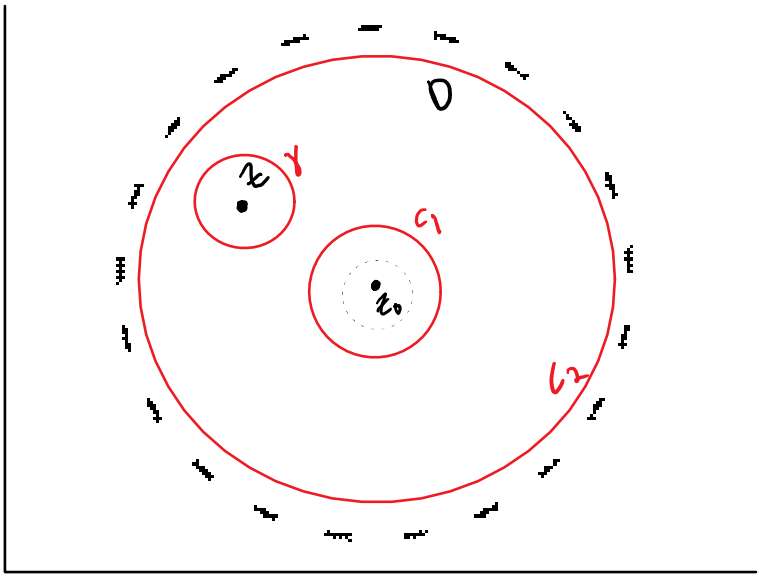
Where

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z - z_0)^{n+1}}dz \qquad n = 0,1,2,\ldots \qquad b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z - z_0)^{-n+1}}dz \quad n = 1,2,\ldots.$$

To prove this theorem we will first show it for $z_0 = 0$. Choose an arbitrary z in the annulus where f is analytic and cleverly re-write $f(z)$ interms of some contour integrals which will give us our Laurent series.



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Notice that this is a multiply connected domain, so if we were to orient C_2 positively and γ, C_1 negatively we have the following theorem that will apply:

Suppose that C is a simple closed contour positively oriented

1. C is a simple closed contour which is positively oriented
2. C_k $(k = 1,2, \dots n)$ are simpleclosed contours interior to C , all negatively oriented(clockwise) and disjoint with no comon interior points.

Then if $f(z)$ is analytic on all of these contours and throughout the multiply connected domain consisting of the points inside C and exterior to each C_k then

$$\int_C f(z)dz + \sum_{k=1}^{n} \int_{c_k} f(z)dz = 0$$

What we will do is define $g(z) = \frac{f(z)}{s-z}$ which is analytic on the multiply connected domain and further more $f(z)$ is analytic on and inside the region defined by γ

We can then apply our theorem $\int_{C_1} g(z)ds + \int_{c_2} g(z)\,ds + \int_{\gamma} g(z)ds = 0$ or interms of $f(s)$

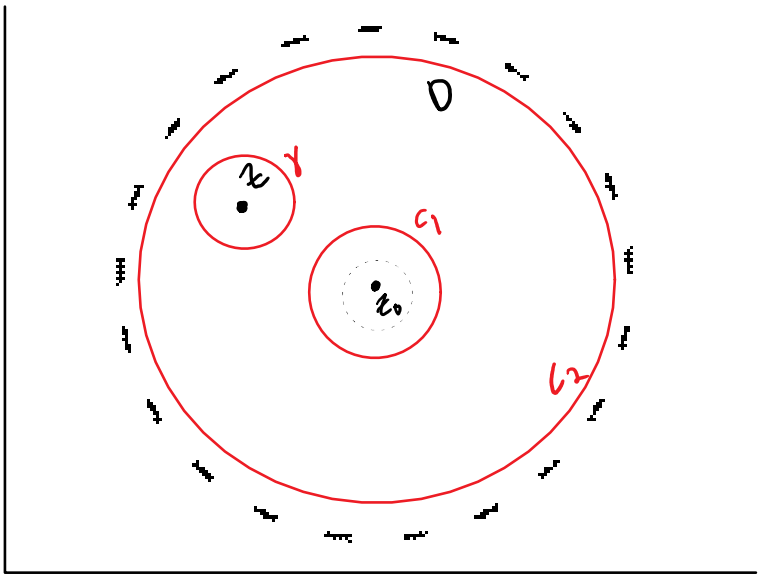
$$\int_{C_1} \frac{f(s)}{s-z}ds + \int_{c_2} \frac{f(s)}{s-z}ds + \int_{\gamma} \frac{f(s)}{s-z}ds = 0$$

Note that if instead we orient C_2 and γ_2 positively we can just change the signs on their integrals that is we can say:

$$\int_{C_1} \frac{f(s)}{s-z}ds - \int_{c_2} \frac{f(s)}{s-z}ds - \int_{\gamma} \frac{f(s)}{s-z}ds = 0$$

Where C_2 and γ anre now positively oriented, that is we just used the fact that $\int_C f(z)dz = -\int_{-C} f(z)dz$

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Ok so we have the statement that :

$$\int_{C_1} \frac{f(s)}{s-z} ds - \int_{c_2} \frac{f(s)}{s-z} ds - \int_{\gamma} \frac{f(s)}{s-z} ds = 0$$

Now the only point where $f(s)$ would not be analytic Is at z_0 this means it is analytic on and in the region defined by γ which means that the cauchy integral formula applies for that contour.

$$\int_{\gamma} \frac{f(s)}{s-z} ds = 2\pi i f(z)$$

So we may say $$\int_{C_1} \frac{f(s)}{s-z} ds - \int_{c_2} \frac{f(s)}{s-z} ds - 2\pi i f(z) = 0$$

Or

$$f(z) = \frac{1}{2\pi i} \int_{C_1} \frac{f(s)}{s-z} ds - \frac{1}{2\pi i} \int_{c_2} \frac{f(s)}{s-z} ds$$

Now we will just multiply that minus sign through the second integral to get

$$f(z) = \frac{1}{2\pi i} \int_{C_1} \frac{f(s)}{s-z} ds + \frac{1}{2\pi i} \int_{c_2} \frac{f(s)}{z-s} ds$$

If we then factor out a $\frac{1}{s}$ from the integral around C_1 and a $\frac{1}{z}$ from the integral around C_2 we can say

$$f(z) = \frac{1}{2\pi i} \int_{C_1} \frac{1}{s} \frac{f(s)}{1-\frac{z}{s}} ds + \frac{1}{2\pi i} \int_{c_2} \frac{1}{z} \frac{f(s)}{1-\frac{s}{z}} ds$$

Any ideas on another way we can represent $\frac{1}{1-\frac{z}{s}}$ *and* $\frac{1}{1-\frac{s}{z}}$?

$$\frac{1}{1-\frac{z}{s}} =$$

$$\frac{1}{1-\frac{s}{z}} =$$

So we have:

$$f(z) = \frac{1}{2\pi i}\int_{C_1} \frac{f(s)}{s}\frac{1}{1-\frac{z}{s}}ds + \frac{1}{2\pi i}\int_{c_2} \frac{f(s)}{z}\frac{1}{1-\frac{s}{z}}ds$$

$$= \frac{1}{2\pi i}\int_{C_1} \frac{f(s)}{s}\sum_{n=0}^{\infty}\left(\frac{z}{s}\right)^n ds + \frac{1}{2\pi i}\int_{c_2} \frac{f(s)}{z}\sum_{n=0}^{\infty}\left(\frac{s}{z}\right)^n ds$$

Our goal will be to interchange the summation and the integration, which we can certainly do for finite sums however these are infinite sums so we have to be a bit more careful, the trick will be to re-write our geometric series

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z} - \frac{z^n}{1-z} + \frac{z^n}{1-z} = \frac{1-z^n}{1-z} + \frac{z^n}{(1-z)} = \sum_{n=0}^{N-1} z^n + \frac{z^n}{1-z}$$

If we put this into our integral equation for $f(z)$ we get:

$$f(z) = \frac{1}{2\pi i}\int_{C_1} \frac{f(s)}{s}\sum_{n=0}^{N-1}\left(\frac{z}{s}\right)^n + \frac{1}{2\pi i}\frac{\left(\frac{z}{s}\right)^n}{1-\frac{z}{s}}ds + \frac{1}{2\pi i}\int_{c_2} \frac{f(s)}{z}\sum_{n=0}^{N-1}\left(\frac{s}{z}\right)^n + \frac{1}{2\pi i}\frac{\left(\frac{s}{z}\right)^n}{1-\frac{s}{z}}ds$$

Now that we have a finite sum we can interchange the summation with the integration:

So we have:

$$f(z)=\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_1}\frac{f(s)}{s^{n+1}}ds\,z^n+\frac{1}{2\pi i}\int_{C_1}\frac{\left(\frac{z}{s}\right)^N}{1-\frac{z}{s}}ds+\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_2}f(s)s^n ds\;\frac{1}{z^{n+1}}+\frac{1}{2\pi i}\int_{C_2}\frac{\left(\frac{s}{z}\right)^N}{1-\frac{s}{z}}ds$$

If we re-index $\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_2}f(s)s^n ds\;\frac{1}{z^{n+1}}$ by starring at $n=1$ and sue negative exponents with s we can write

$$\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_2}f(s)s^n ds\;\frac{1}{z^{n+1}}=\sum_{n=1}^{N}\frac{1}{2\pi i}\int_{C_2}\frac{f(s)}{s^{-n+1}}ds\;\frac{1}{z^n}$$

So we now have that

$$f(z)=\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_1}\frac{f(s)}{s^{n+1}}ds\,z^n+\frac{1}{2\pi i}\int_{C_1}\frac{\left(\frac{z}{s}\right)^N}{1-\frac{z}{s}}ds+\sum_{n=1}^{N}\frac{1}{2\pi i}\int_{C_2}\frac{f(s)}{s^{-n+1}}ds\;\frac{1}{z^n}+\frac{1}{2\pi i}\int_{C_2}\frac{\left(\frac{s}{z}\right)^N}{1-\frac{s}{z}}ds$$

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Now recall that our point z_0 is the orign, so another way I can write this is

$$f(z)=\sum_{n=0}^{N-1}\frac{1}{2\pi i}\int_{C_1}\frac{f(s)}{(s-0)^{n+1}}ds\,(z-0)^n+\frac{1}{2\pi i}\int_{C_1}\frac{\left(\frac{z}{s}\right)^N}{1-\frac{z}{s}}ds+\sum_{n=1}^{N}\frac{1}{2\pi i}\int_{C_2}\frac{f(s)}{(s-0)^{-n+1}}ds\,\frac{1}{(z-0)^n}+\frac{1}{2\pi i}\int_{C_2}\frac{\left(\frac{s}{z}\right)^N}{1-\frac{s}{z}}ds$$

Now we take the limit as $N\to\infty$ it turns out that both

$$\lim_{N\to\infty}\frac{1}{2\pi i}\int_{C_1}\frac{\left(\frac{z}{s}\right)^N}{1-\frac{z}{s}}ds=0 \qquad \text{And} \qquad \lim_{N\to\infty}\frac{1}{2\pi i}\int_{C_2}\frac{\left(\frac{s}{z}\right)^N}{1-\frac{s}{z}}ds=0$$

This means in our limit we obtain

$$f(z)=\sum_{n=0}^{\infty}\frac{1}{2\pi i}\int_{C_1}\frac{f(s)}{(s-0)^{n+1}}ds\,(z-0)^n+\sum_{n=1}^{\infty}\frac{1}{2\pi i}\int_{C_2}\frac{f(s)}{(s-0)^{-n+1}}ds\,\frac{1}{(z-0)^n}$$

Compare this series to the Laurent series coefficients if $z_0=0$

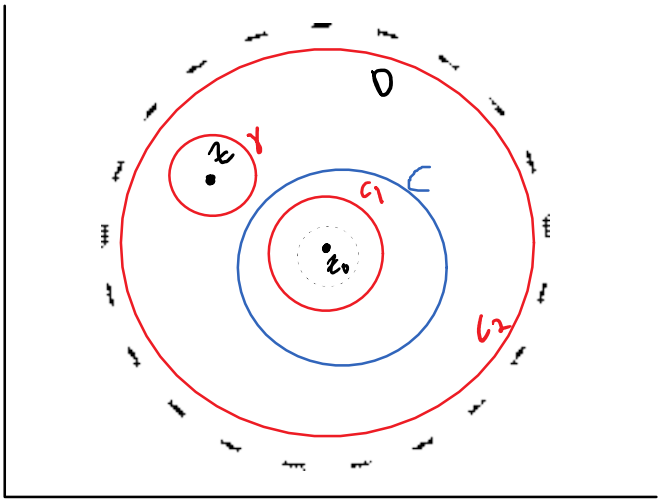
$$f(z)=\sum_{n=0}^{\infty}a_n(z-z_0)^n+\sum_{n=1}^{\infty}\frac{b_n}{(z-z_0)^n}\qquad R_1<|z-z_0|<R_2$$

.

Where

$$a_n=\frac{1}{2\pi i}\int_C\frac{f(z)}{(z-z_0)^{n+1}}dz \qquad n=0,1,2,\ldots \qquad b_n=\frac{1}{2\pi i}\int_C\frac{f(z)}{(z-z_0)^{-n+1}}dz \quad n=1,2,\ldots.$$

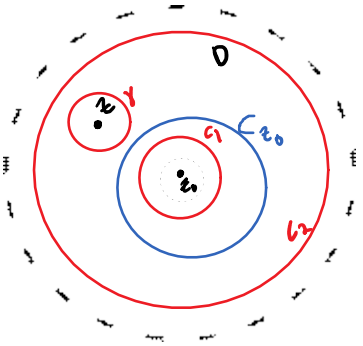
We need to keep in mind our deformation theorems, let C be some contour in between C_1, C_2

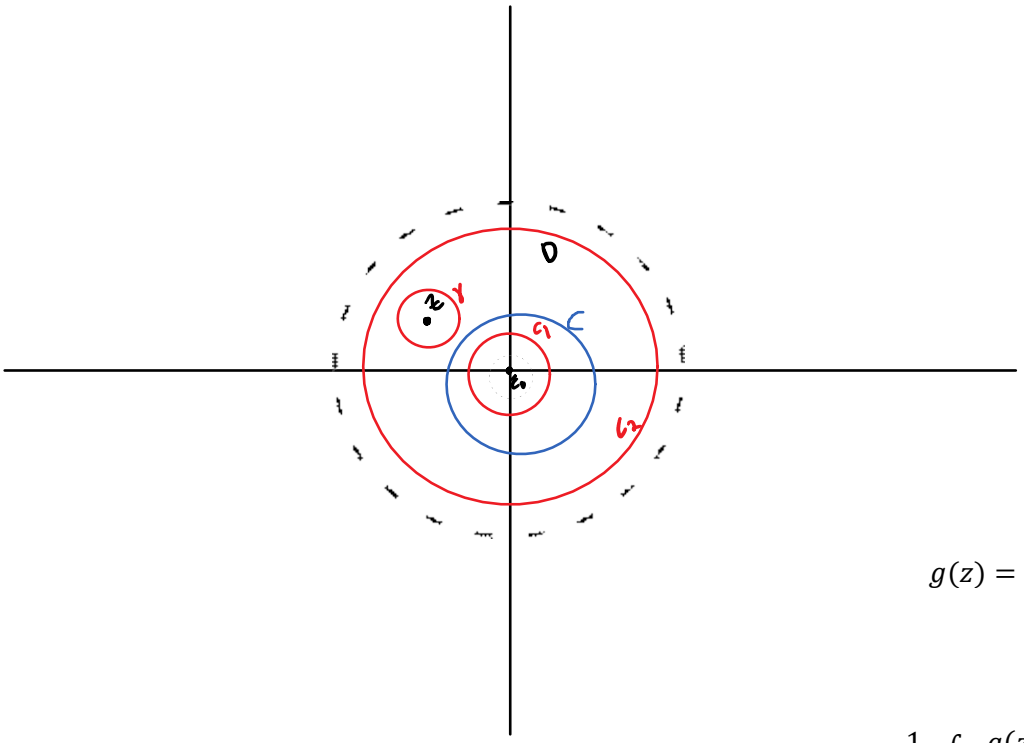


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So we found our Laurent series when $f(z)$ is not analytic at the orign!

To extend it to the case for any z_0 we simply define $g(z) = f(z + z_0)$ if $f(z)$ is analytic on $R_1 < |z - z_0| < R_2$ then $g(z)$ is on $R_1 < |z + z_0 - z_0| < R_2 \rightarrow R_1 < |z| < R_2$ which is centered at the orign now. Also the contours for $f(z)$ can be shifted by z_0 to be around the orign that is $C = z(t) - z_0$





$$g(z) = \sum_{n=0}^{\infty} a_n (z)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z)^n}$$

$$a_n = \frac{1}{2\pi i} \int_C \frac{g(z)}{(z)^{n+1}} dz \qquad b_n = \frac{1}{2\pi i} \int_C \frac{g(z)}{(z)^{-n+1}} dz$$

This means what we just proves applies to $g(z)$ that is:

But $g(z) = f(z + z_0) = \sum_{n=0}^{\infty} a_n (z)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z)^n}$ $f(z) = f(z + z_0 - z_0) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$

and

$$a_n = \frac{1}{2\pi i} \int_C \frac{g(z)}{(z)^{n+1}} dz = \int_a^b \frac{g(z(t) - z_0)}{(z(t) - z_0)^{n+1}} z'(t) dt = \int_a^b \frac{f(z(t) - z_0 + z_0)}{(z(t) - z_0)^{n+1}} z'(t) dt = \int_a^b \frac{f(z(t))}{(z(t) - z_0)^{n+1}} z'(t) dt = \int_{C_{z_0}} \frac{f(z)}{(z - z_0)^{n+1}} dz$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{g(z)}{(z)^{-n+1}} dz = \int_a^b \frac{g(z(t) - z_0)}{(z(t) - z_0)^{-n+1}} z'(t) dt = \int_a^b \frac{f(z(t) - z_0 + z_0)}{(z(t) - z_0)^{-n+1}} z'(t) dt = \int_a^b \frac{f(z(t))}{(z(t) - z_0)^{-n+1}} z'(t) dt = \int_{C_{z_0}} \frac{f(z)}{(z - z_0)^{-n+1}} dz$$

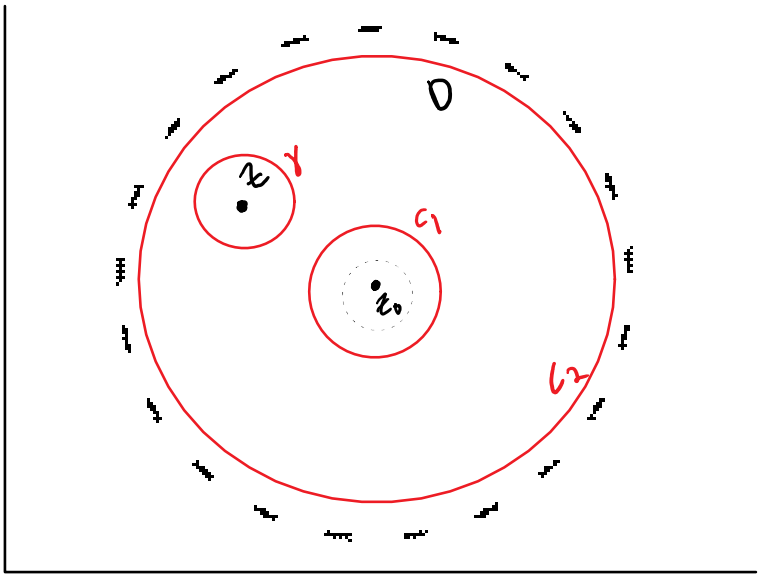
And there we go!

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Now we can very easily prove Taylor's theorem

Recall that if $g(z)$ is analytic on and in the region Defined by a simple closed contour

$$\int_C g(z)dz = 0$$



Well suppose our $f(z)$ is analytic at z_0 and consider

$$b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{-n+1}}dz \quad n = 1,2,\ldots.$$

$$b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{-n+1}}dz = \frac{1}{2\pi i}\int_C f(z)(z-z_0)^{n-1}dz$$

Note then that $g(z) = f(z)(z-z_0)^{n-1}$ is analytic on the contour around z_0 and inside since z_0 is not a problem here as the denomonitor comes up on top. This means that $b_n = 0$

If that is the case then we only retain the part of our Laurent series with the $a_n's$

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0) + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$$

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}dz \qquad n = 0,1,2,\ldots$$

$$b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{-n+1}}dz \quad n = 1,2,\ldots.$$

Then since $f(z)$ is analytic at z_0 we can applu the extended cauchy integral formula

$$\frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}dz = \frac{1}{n!}f^{(n)}(z_0) = \qquad n = 0,1,2,\ldots$$

Thus

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{n+1}}dz = \frac{f^n(z_0)}{n!}$$

Our Taylor Series Coefficients!!!!

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Homework Example

Derive the Laurent series representation of

$$f(z) = \frac{e^z}{(z+1)^2}$$

about the point $z_0 = -1$.

Let's also just verify that $b_n = 0$ for $n > 2$ with the integral definition of the cofficent $b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z-z_0)^{-n+1}}\,dz$

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Homework Example

On the domain $0 < |z-1| < 2$, find the Laurent series about $z_0 = 1$ of

$$f(z) = \frac{z}{(z-1)(z-3)}$$

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Now just a couple of statements on term by term differentiation and contour integration of power series.

Suppose we have a power series:

$$S(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Theorem: Let C denote any contour interior to the circle of convergence of a powerseries, and let $g(z)$ be any function that in continuous on C. The series formed by multiplying the power series by $g(z)$ can be integrated term by term over C

$$\int_C g(s)S(z)dz = \sum_{n=0}^{\infty} a_n \int_C g(z)\,(z - z_0)^n dz$$

Note that the function $g(z) = 1$ is continuious on any contour.

Theorem: The power series can be differentiated term by term, that is at each point z interior to the circle of convergence of that series

$$S'(z) = \sum_{n=0}^{\infty} n a_n (z - z_o)^{n-1}$$

A statement on Uniqueness

Theorem: If a series

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$$

Converges to $f(z)$ at all points in some annular domain about z_0, Then it is the Laurant series expansion for f in powers of $z - z_0$ in that domain, if $f(z)$ is analytic in the whole disk $|z - z_0| < R$ then $b_n = 0$ and the weries is the taylor series.

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Homework Example

Find the Taylor Series for $f(z) = \frac{1}{(1-z)^2}$ for the disk $|z| < 1$